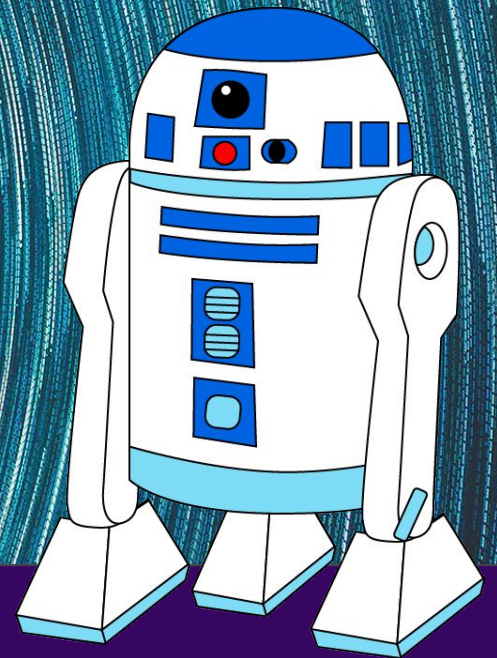


CIS 421/521:
ARTIFICIAL INTELLIGENCE

Machine Learning with Perceptrons



Learning Agents

Machine Learning =
a computer observes data
builds a model
uses the model as a hypothesis about the world
and as software that can solve problems

In reinforcement learning, we studied agents that could make **observations** about the world and **learn** from them. They built a **model** based on the data that they collected and used that to optimize their policy.

Learning Agents

Many of an agent's components could be improved with machine learning:

1. Mapping from conditions on the current state to actions
2. Inferring relevant properties of the world from percepts
3. What will be the result of taking an action?

**Learn by trying actions and observing effects.
Breaking on a wet road.**

By being shown many pictures labeled as bicycles, it should learn to recognize them.

By observing human drivers, a self driving car should learn under what conditions it should break.

Supervised Machine Learning

In supervised learning the agent is given pairs of inputs and outputs, and learns a function to map from the input to the output.

The input is typically a **factored representation** like a vector of values.

In **classification problems** the output is one of a finite set of values like *true/false* or *sunny/cloudy/rainy*.

In **regression problems** the output can be any number, like tomorrow's temperature.

Supervised Learning

Given a training set of N example input-output pairs

$$(x_1, y_1), (x_2, y_2), \dots (x_N, y_N)$$

where each pair was generated by an underlying unknown function $y = f(x)$

the learner should discover a function h that approximates the true function f .